HW1: Three-Layer NumPy MLP for EuroSAT Land-Cover Classification

1. Task and Dataset

This work implements a multilayer perceptron classifier from scratch with NumPy only. The model is trained on EuroSAT_RGB, a 10-class remote-sensing land-cover dataset. Images are resized to 32x32 RGB, flattened, scaled to [0, 1], and standardized using the training-set mean and standard deviation.

The deterministic stratified split is train/validation/test = 18900/4050/4050.

2. Model and Optimization

The network uses fully-connected layers input -> hidden1 -> hidden2 -> softmax output. Forward propagation, back propagation, softmax cross-entropy, L2 weight decay, SGD/momentum updates, and learning-rate decay are all implemented manually. The validation-set accuracy is used to save the best model weights during training.

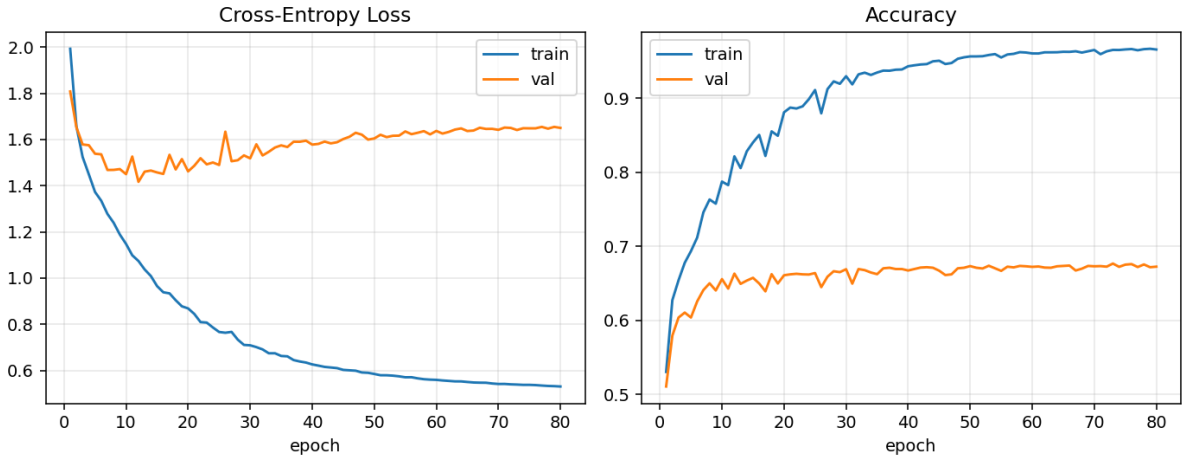
Item	Value
Best run	1
Hidden dimensions	[384, 96]
Activation	relu
Initial learning rate	0.01
Learning-rate decay	0.97
L2 weight decay	0.001
Momentum	0.9
Best validation accuracy	67.68%
Test accuracy	68.27%

3. Hyperparameter Search

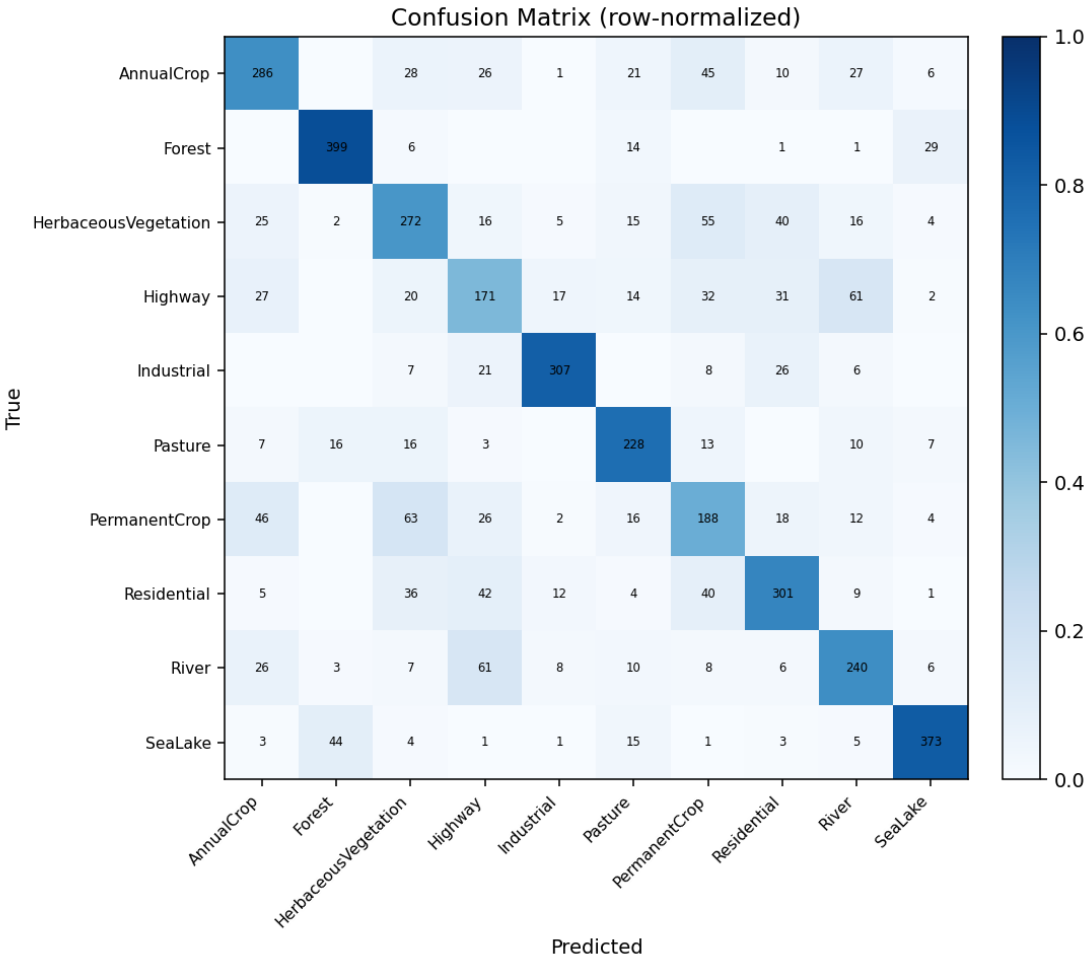
Run	Hidden	Act.	LR	Decay	L2	Mom.	Best Val. Acc.
1	[384, 96]	relu	0.01	0.97	0.001	0.9	67.68%
2	[256, 64]	relu	0.012	0.97	0.001	0.9	67.33%
3	[256, 64]	relu	0.012	0.97	0.0007	0.9	67.01%
4	[256, 64]	relu	0.035	0.95	0.0007	0.0	66.91%
5	[256, 128]	tanh	0.015	0.95	0.0001	0.0	58.59%

4. Training Curves and Test Evaluation

Loss and Accuracy Curves

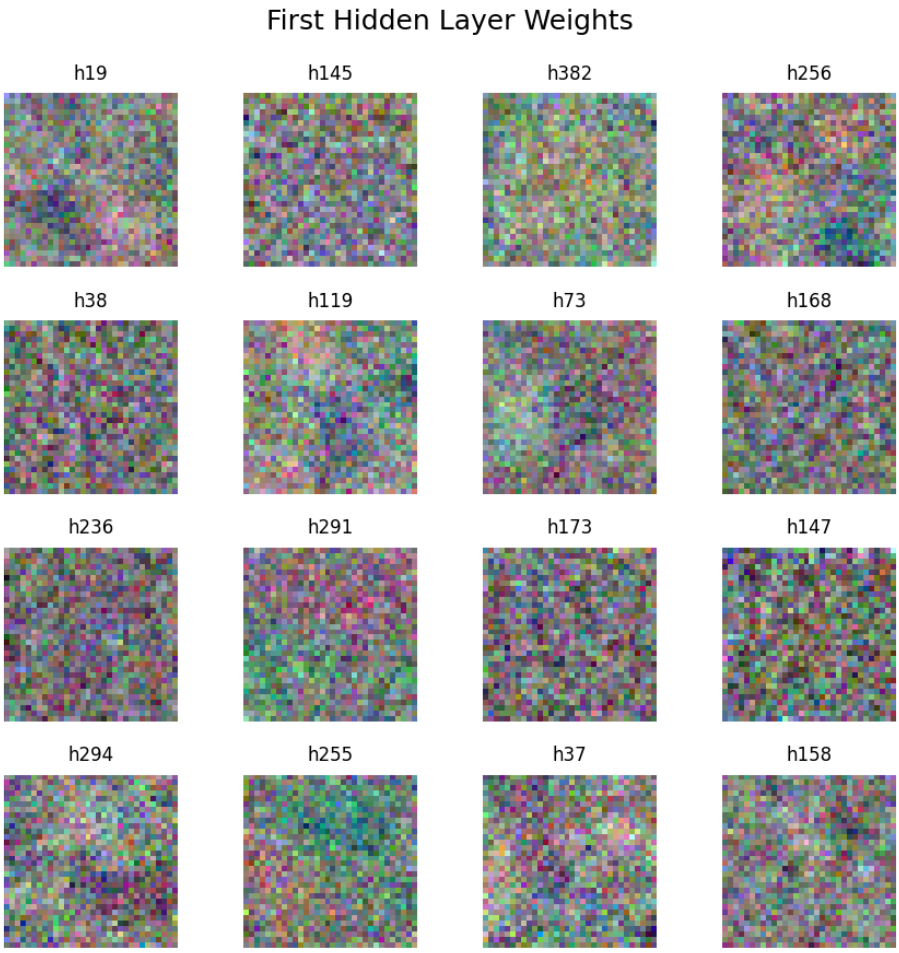


Confusion Matrix



The confusion matrix shows that visually similar categories are the main source of errors. For example, River and Highway both contain elongated linear structures, while AnnualCrop, PermanentCrop, Pasture, and HerbaceousVegetation often share green/brown field textures. The simple MLP has no convolutional inductive bias, so it must learn spatial patterns only from flattened pixels; this explains why texture-rich or layout-dependent classes remain hard.

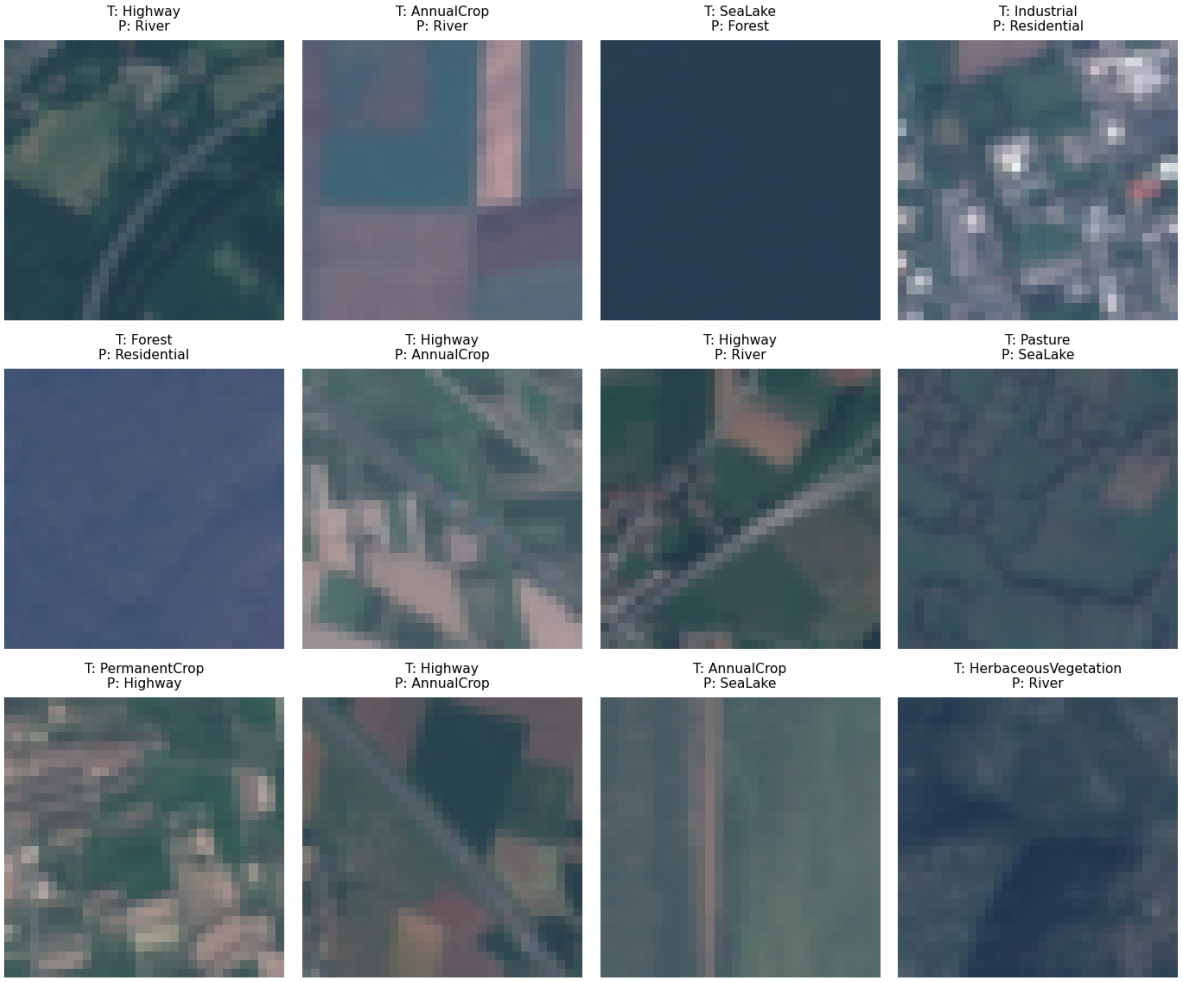
5. First-Layer Weight Visualization



The first hidden-layer weights are reshaped back to RGB image grids. The stronger filters show broad color and low-frequency spatial preferences instead of sharp local edge detectors. This is expected for a fully-connected classifier trained on small satellite thumbnails: early units tend to capture global vegetation, water, soil, and urban color patterns.

6. Error Analysis

Misclassified Test Examples



The sampled mistakes confirm the quantitative pattern. Highways can be confused with rivers because both appear as thin continuous paths. Cropland, pasture, and herbaceous vegetation also overlap when the image contains mixed land parcels or lacks distinctive structure. A convolutional model would likely improve these cases, but the assignment intentionally uses an MLP to practice manual differentiation and optimization.

7. Reproducibility

Run `python3 src/train_numpy_mlp.py --data-dir EuroSAT_RGB --out-dir outputs --search` from the homework directory to reproduce the training and figures. Run `python3 src/make_report.py` to regenerate this PDF. The saved model weights are in `outputs/best_model.npz`.

Public GitHub Repo	https://github.com/huang200309/eurosat-numpy-mlp-hw1
Model Weights Download	https://github.com/huang200309/eurosat-numpy-mlp-hw1/raw/main/outputs/best_model.np